



Asset Performance Management

System Specifications Guide

Release 540.1

August 2023

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CAPACITY

The Asset Performance Management (APM) application is used in different scenarios and plants of differing sizes. Consequently, the capacity parameters for APM depend upon the size of these plants. There are various factors that define the sizing of an APM system deployment. These factors vary from one plant to another. Hence, it is difficult to define one fixed system sizing for APM.

The following sections specify the APM system sizing based on the size of the plant on a scale of - small, medium, and large plant. To define the size for these systems the following parameters are used:

- Number of assets/attributes/properties in a plant.
- Number of calculations involved, frequency, and processing time.
- Number of concurrent users.

NOTE

These sections denote an indicative number of servers/VMs required. For every site, the projects team should further determine the sizing parameters apart from the above like - frequency of change of inputs, complexity of calculations, frequency of calculations, model of equipment etc. and test the deployment taking the below specifications as a base to ensure required system performance. As these sizing parameters will differ from one plant to another, it may be possible that the number of servers required may increase or decrease. Hence, actual number of servers required will totally depend on the results of testing.

LARGE SYSTEM SPECIFICATIONS

2.1 Capacity Parameters

Below table list the capacity parameters for a large sized plant:

Parameters	Maximum Limit
# of Assets	Up to 40k asset up to 2lac attributes and 50 lac properties
# of Calculations	200000(20% - 1 minute, 30% - 2 minute, 50% - 5 minute)
	Scheduled - 25k
Inputs frequency	>= 2 second
Outputs	Upto 40000 per minute per calc server
Calc processing time	< 5 millisecond
# of Concurrent Users	Up to 100

2.2 Hardware and Software Specifications

Below table lists the physical server\virtual nodes needed for Asset Performance Management as applicable to the capacity parameters listed in the table above:

7 Box PCT - Hardware and Software Specification					
Sr. No.	Server	RAM	Processor	HDD Size	Software
1	Web Server	32 GB	4 Quad x 3.3GHz	500 GB	- OS: Windows 2016 or Windows 2019 - Microsoft.Net Framework 4.8 - Microsoft.NET Framework 3.5 Service pack 1 - Microsoft Office 2013 or Microsoft Office 2016 (64bit) or Office 365
2	App Server	32 GB	4 Quad x 3.3GHz	500 GB	.OS: Windows 2016 or Windows 2019 - Microsoft.Net Framework 4.8 - Microsoft.NET Framework 3.5 Service pack 1
3	Database Server	32 GB	4 Quad x 3.3GHz	500 GB	- SQL 2016 / 2017 / 2019 Standard or Enterprise Edition (64 bit) - InfluxDB 1.8)
4	Calc1 Server	64 GB	16 Quad x 3.3GHz	500 GB	OS: Windows 2016 or Windows 2019 .Net Framework 4.8

7 Box PCT - Hardware and Software Specification					
Sr. No.	Server	RAM	Processor	HDD Size	Software
5	Calc2 Server	64 GB	16 Quad x 3.3GHz	500 GB	OS: Windows 2016 or Windows 2019 .Net Framework 4.8
6	Calc3 Server	64 GB	16 Quad x 3.3GHz	500 GB	OS: Windows 2016 or Windows 2019 .Net Framework 4.8
7	Calc4 Server	64 GB	16 Quad x 3.3GHz	500 GB	OS: Windows 2016 or Windows 2019 .Net Framework 4.8

MEDIUM SYSTEM SPECIFICATIONS

3.1 Capacity Parameters

Parameters	Maximum Limit
# of Assets	Up to 10k assets
# of Calculations	40000(20% - 1 minute, 30% - 2 minute, 50% - 5 minute) with a max number of 2000 thermo & embedded calculations per minute Scheduled - 11552 Data Driven - 5000 Fault model -24435
Inputs frequency	>= 5 minutes
Outputs	Upto 10000 per minute per calc server
Calc processing time	< 100 milliseconds
# of Concurrent Users	Up to 50

3.2 Capacity Parameter with Analytics

Parameters	Maximum Limit
# of Python Calculations	Total calculations: 300 (10% - 1 minute, 20% - 5 minute, 70% - 30 minutes) Scheduled - 25 On Demand - 175
Input Frequency	>= 1 minutes
Outputs	1 lac per 5 minutes
Calc processing time	< 5 minutes with max 1 lac data points (no.of data points are arrived based on interval and duration and attributes/properties)
# of Concurrent Models	Up to 5 concurrent models
# of Concurrent Users	Up to 5 for model for configuration, up to 10 for runtime

3.3 Physical Nodes Specifications

Below table lists the physical server\virtual nodes needed for APM as applicable to the capacity parameters listed in the table above

5 Box PCT - Hardware and Software Specification					
Sr. No.	Server	RAM	Processor	HDD Size	Software
1	Web Server	32 GB	4 Quad x 3.3GHz	500 GB	- OS: Windows 2016 or Windows 2019 - Microsoft.Net Framework 4.8 - Microsoft.NET Framework 3.5 Service pack 1 - Microsoft Office 2013 or Microsoft Office 2016 (64bit)
2	App Server	32 GB	4 Quad x 3.3GHz	500 GB	- OS: Windows 2016 or Windows 2019 - Microsoft.Net Framework 4.8 - Microsoft.NET Framework 3.5 Service pack 1 - Microsoft Office 2013 or Microsoft Office 2016 (64bit)
3	Database Server	32 GB	4 Quad x 3.3GHz	500 GB	SQL 2016 / 2017 / 2019 Standard or Enterprise Edition (64 bit)
4	Calc1 Server	64 GB	16 Quad x 3.3GHz	500 GB	OS: Windows 2016 or Windows 2019 .Net Framework 4.7.2
5	Calc2 Server	64 GB	16 Quad x 3.3GHz	500 GB	OS: Windows 2016 or Windows 2019 .Net Framework 4.7.2

SMALL SYSTEM SPECIFICATIONS

4.1 Capacity Parameters

Table below lists the capacity parameters for a small sized plant.

Parameters	Maximum Limit
# of Assets	Up to 5k assets
# of Calculations	10000(20% - 1 minute, 30% - 2 minute, 50% - 5 minute) Scheduled - 25k Data Driven - ? Fault model - ?
Inputs frequency	>= 2 second
Outputs	Upto 40000 per minute per calc server
Calc processing time	< 5 millisecond
# of Concurrent Users	Up to 50

Here 20%, 30% and 50% denotes the change in inputs, causing the calculations to be triggered. That means 20% calculations get triggered every one minute, 30% calculations get triggered every 2 minutes and 50% calculations get triggered every 5 minutes.

4.2 Hardware Specification

Below table lists the physical server\virtual nodes needed for APM as applicable to the capacity parameters listed in the above table

2 Box PCT - Hardware and Software Specification					
Sr. No.	Server	RAM	Processor	HDD Size	Software
1	Web/App/Calc Server	32 GB	8 Quad x 3.3GHz	500 GB	- OS: Windows 2016 or Windows 2019 - Microsoft.Net Framework 4.8 - Microsoft.NET Framework 3.5 Service pack 1 - Microsoft Office 2013 or Microsoft Office 2016 (64bit)
2	Database Server	32 GB	4 Quad x 3.3GHz	500 GB	SQL 2016 / 2017 / 2019 Standard or Enterprise Edition (64 bit)

ANALYTICS SYSTEM SPECIFICATIONS - TWO SERVER

5.1 Capacity Parameters

Table below lists the capacity parameters for a Analytics models.

Parameters	Maximum Limit
# of Assets	Up to 100 assets up to 6k attributes and 30k properties
# of Python Calculations	Total calculations: 500 (10% - 1 minute, 20% - 5 minute, 70% - 30 minutes) Scheduled - 300 On Demand - 200
Inputs frequency	>= 1 minute
Outputs	1 lac per 5 minutes
Calc processing time	< 5 minutes with max 1 lac data points (no.of data points are arrived based on interval and duration and attributes/properties)
# of Concurrent Models	Up to 5 concurrent models
# of Concurrent Users	Up to 5 for model for configuration, up to 10 for runtime

5.2 Hardware Specifications

Below table lists the physical server\virtual nodes needed for APM as applicable to the capacity parameters listed in Table 1.

2 Box PCT - Hardware and Software Specification					
Sr. No.	Server	RAM	Processor	HDD Size	Software
1	Web/App/Calc Server	32 GB	8 Quad x 3.3GHz	500 GB	- OS: Windows 2016 or Windows 2019 - Microsoft.Net Framework 4.8 - Microsoft.NET Framework 3.5 Service pack 1 - Microsoft Office 2013 or Microsoft Office 2016 (64bit)
2	Database Server	32 GB	4 Quad x 3.3GHz	500 GB	- SQL 2016 / 2017 / 2019 Standard or Enterprise Edition (64 bit) - InfluxDB 1.8

ANALYTICS SYSTEM SPECIFICATIONS - SINGLE SERVER

6.1 Capacity Parameters

Table below lists the capacity parameters for a Analytics models.

Parameters	Maximum Limit
# of Assets	Up to 50 assets up to 2k attributes and 10k properties
# of Python Calculations	Total calculations: 200 (10% - 1 minute, 20% - 5 minute, 70% - 30 minutes) Scheduled - 100 On Demand - 100
Inputs frequency	>= 1 minute
Outputs	50k per 5 minutes
Calc processing time	< 5 minutes with max 50k data points (no.of data points are arrived based on interval and duration and attributes/properties)
# of Concurrent Models	Up to 2 concurrent models
# of Concurrent Users	Up to 2 for model for configuration, up to 5 for runtime

6.2 Hardware Specifications

Below table lists the physical server\virtual nodes needed for APM as applicable to the capacity parameters listed in Table 1.

2 Box PCT - Hardware and Software Specification					
Sr. No.	Server	RAM	Processor	HDD Size	Software
1	Web/App/Calc Server/ DB Server	32 GB	8 Quad x 3.3GHz	500 GB	- OS: Windows 2016 or Windows 2019 - Microsoft.Net Framework 4.8 - Microsoft.NET Framework 3.5 Service pack 1 - Microsoft Office 2013 or Microsoft Office 2016 (64bit) - InfluxDB 1.8

CLIENT REQUIREMENT

(This can be still in the main document, may not be required here)

Sr. No.	Server	Client Requirement			
		RAM	Processor	HDD Size	Software
1	Client (plain client without any Asset Performance Management tools)	32 GB	4 Quad x 3.3GHz	500 GB	- Windows 8.1 (32bit and 64bit) or Windows 10 (32bit and 64bit) - Internet Explorer 11.0.9600.18537 (KB3203621) - Microsoft.NET Framework 3.5 Service pack 1 .Microsoft.NET Framework 4.8 - Google Chrome 74.0.3729.108 and above
2	Asset Performance Management Synchronizer Requirement	32 GB	4 Quad x 3.3GHz	500 GB	- Windows 8.1 (32bit) or Windows 10 (32bit) - Experion Flex Station - Microsoft.NET Framework 3.5 Service pack 1 .Microsoft.NET Framework 4.8
3	APM Configuration Excel Plugin	32 GB	4 Quad x 3.3GHz	500 GB	- Windows 8.1 (32bit) or Windows 10 (64bit) - Microsoft Office 2013 or Microsoft Office 2016 (64bit) - Microsoft Office must have been installed using the "All Users" option in the installation wizard.

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